



Electrical and Computer Engineering
University of Thessaly

ECE214 - Introduction to Electronics

Characterization of CMOS Inverter Delay and Interconnects in SPICE

Spring Semester — 2021-2022

Assignment 3
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Submission Date: May 1, 2022

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Exercise 3.1

1 Delay Analysis of a CMOS Inverter

In this section, we study the transient response of a CMOS inverter. Specifically, we analyze how the propagation delays t_{pHL} and t_{pLH} are affected by the sizing of the NMOS and PMOS transistors. The measurements are performed at the 50% threshold of the input and output voltage waveforms.

In the context of this study, the ratio $\frac{W_p}{W_n} = \frac{k'_n}{k'_p} = \frac{2.5 \cdot 10^{-4}}{5.194 \cdot 10^{-5}} = 4.813$ is maintained constant. Simultaneously, by continuously adjusting the channel width of the NMOS transistor across the values $W_n \in \{2 \mu\text{m}, 3.2 \mu\text{m}, 6 \mu\text{m}, 10 \mu\text{m}, 20 \mu\text{m}\}$, while ensuring that the area and perimeter of the drain and source regions are scaled accordingly, the simulation waveforms shown in Figure ?? are obtained. Figure ?? illustrates the transient response of the inverter for different combinations of values W_n and W_p to ensure that their ratio remains stable.

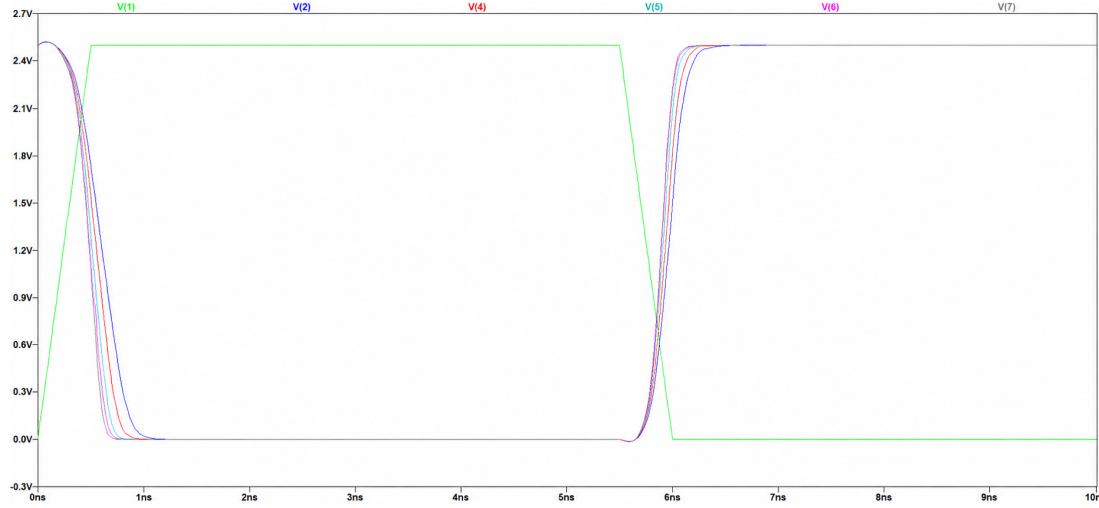


Figure 1: Transient Response of the Inverter for different values of W_n, W_p

It is observed that as the channel width (W_n) of the various inverters increases, the propagation delay for both the rise and fall transitions decreases. Concurrently, a diminishing marginal return (a drop in the reduction rate) is noted regarding these delays. Specifically, the high-to-low propagation delay is governed by the relation:

$$t_{pHL} = R_n \cdot C_L \cdot \ln(2)$$

By expanding the equivalent large-signal channel resistance, the expression unfolds as follows:

$$t_{pHL} = \frac{2L}{k'_n W_n V_{DD} \ln(2) \left[\frac{7}{4} - \frac{3V_{Tn}}{V_{DD}} + \left(\frac{V_{Tn}}{V_{DD}} \right)^2 \right]} \cdot C_L \cdot \ln(2)$$

Which can be structurally simplified to isolate the geometric parameters:

$$t_{pHL} = \frac{L}{W_n} \cdot \left(\frac{2}{k'_n V_{DD} \left[\frac{7}{4} - \frac{3V_{Tn}}{V_{DD}} + \left(\frac{V_{Tn}}{V_{DD}} \right)^2 \right]} \right) \cdot C_L \cdot \ln(2)$$

Based on the derived analytical formulation, it is concluded that t_{pHL} depends strictly on the transistor width W_n and the load capacitance C_L , given that all other architectural and process parameters for the specific technology node remain constant. It is observed that when W_n scales up, the fraction $\frac{L}{W_n}$ decreases; however, the overall propagation delay does not degrade in a perfectly inversely proportional manner relative to the width. According to the simulation measurements, this non-linear behavior occurs because an increase in W_n inherently introduces an expansion in the device's internal parasitic junction dimensions. This growth scales up the self-loading component of C_L (though not significantly enough to completely counteract the delay optimization). Consequently, the overall rate of delay reduction gradually diminishes as the transistor width scales to higher values.

Similarly, it is derived that the low-to-high (rise) propagation delay (t_{pLH}) is governed by the following analytical formulation:

$$t_{pLH} = R_p \cdot C_L \cdot \ln(2)$$

By expanding the equivalent large-signal channel resistance of the pMOS network, the expression unfolds as:

$$t_{pLH} = \left(\frac{2L}{k'_p W_p V_{DD} \ln(2) \left[\frac{7}{4} - \frac{3|V_{Tp}|}{V_{DD}} + \left(\frac{|V_{Tp}|}{V_{DD}} \right)^2 \right]} \right) \cdot C_L \cdot \ln(2)$$

This expression demonstrates that the rise delay depends fundamentally on the geometric width of the pMOS device (W_p), which is scaled proportionally to the NMOS width to achieve symmetric driving strengths, as well as the cumulative load capacitance (C_L). Through precise simulation measurements, it is observed once again that the rate of delay reduction diminishes as W_p increases. This non-linear behavior is structurally attributed to two competing factors: while scaling up the layout width directly reduces the geometric scaling fraction, it simultaneously expands the internal parasitic diffusion junctions of the device. This parasitic expansion increases the self-loading component of C_L . Although this capacitance growth is insufficient to cause an absolute increase in propagation delay, it actively dampens and minimizes the rate of delay optimization across expanding transistor dimensions.

Exercise 3.2

In this experiment, the dynamic behavioral response and propagation delay of a baseline CMOS inverter (Inverter 1) are evaluated under varying loading constraints. Inverter 1 drives a secondary downstream inverter (Inverter 2), which acts as a purely capacitive fan-out load. The geometric parameters (channel widths W_n and W_p) of Inverter 1 remain constant throughout the evaluation, whereas the widths of Inverter 2 are scaled progressively from $1\times$ to $20\times$ its baseline sizing.

Since the interconnect parasitic capacitances are assumed to be negligible, the total load capacitance (C_L) driven by the output node V_{o1} of the first inverter is equal to the input gate capacitance (C_{g2}) of the second inverter:

$$C_L = C_{g2} \quad (1)$$

The gate capacitance of a MOSFET is directly proportional to its gate area ($C_g = \frac{e_{ox}}{t_{ox}} \cdot W \cdot L$). Given that the channel lengths (L) and e_{ox}, t_{ox} are kept invariant, the gate capacitance scales linearly with the transistor width (W). Therefore, the input capacitance ratio of the two blocks simplifies directly to the geometric scaling factor of Inverter 2:

$$\frac{C_{g(\text{Inverter 2})}}{C_{g(\text{Inverter 1})}} = \frac{W_{\text{Inverter 2}}}{W_{\text{Inverter 1}}} = \{1, 2, 3, \dots, 20\} \quad (2)$$

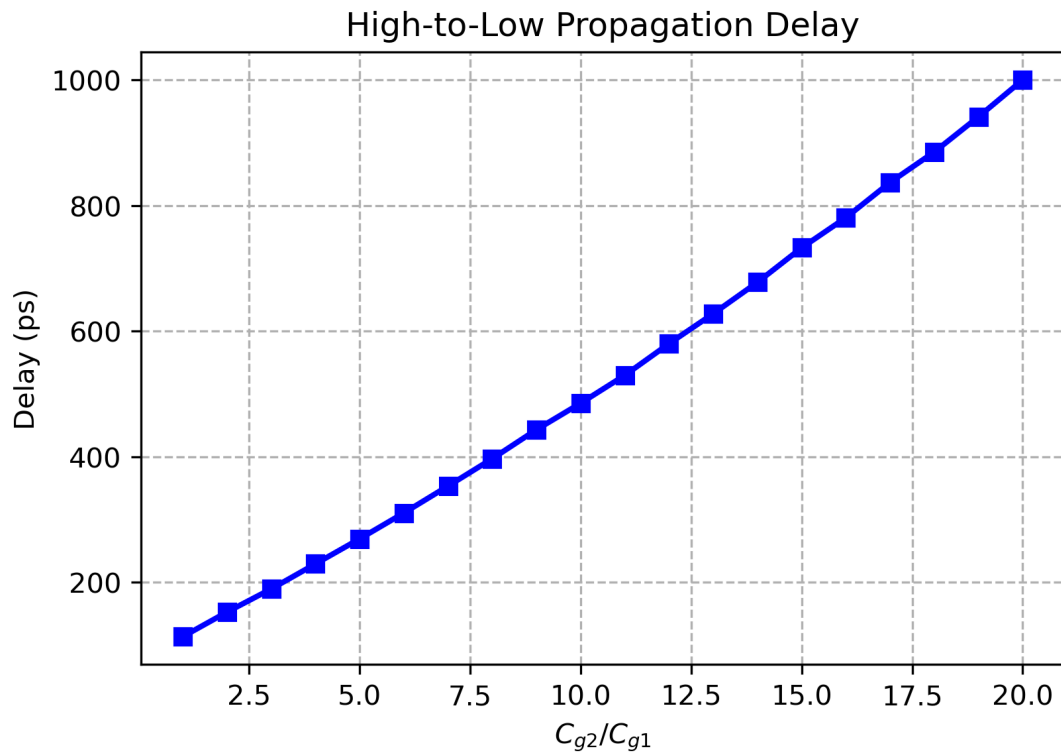
An ideal input step pulse with instantaneous (zero) rise and fall times is applied at V_{in} to isolate the propagation delays from any input edge-rate dependencies.

The transient response of the cascaded structure was simulated 20 consecutive times using the LTspice tool. The extracted high-to-low propagation delay (t_{pHL}) and low-to-high propagation delay (t_{pLH}) between the input of the first inverter and the output of the second inverter are compiled in Table 1.

The graphical representations of both propagation delays t_{pHL} and t_{pLH} with respect to the ratio $\frac{C_{L(ext)}}{C_{in}}$ are illustrated in Figures 2 and Figure 3, respectively.

Table 1: Extracted Propagation Delays for $C_{g(2)}/C_{g(1)}$ Scaling Factors

Scaling Factor of Inverter 2	Capacitance Ratio $C_{g(2)}/C_{g(1)}$	High-to-Low Delay t_{pHL} (ps)	Low-to-High Delay t_{pLH} (ps)
1×	1	113.50	51.60
2×	2	152.67	73.16
3×	3	189.48	91.51
4×	4	229.32	114.10
5×	5	269.22	135.80
6×	6	310.44	156.88
7×	7	353.30	179.20
8×	8	396.86	200.80
9×	9	443.10	222.60
10×	10	485.22	245.07
11×	11	529.70	267.29
12×	12	580.10	290.20
13×	13	627.80	310.00
14×	14	677.80	330.00
15×	15	732.70	355.00
16×	16	780.70	380.00
17×	17	836.80	402.10
18×	18	885.30	421.10
19×	19	941.30	444.38
20×	20	1000.00	468.63

Figure 2: High-to-Low propagation delay (t_{pHL}) with respect to $\frac{C_{L(ext)}}{C_{in}}$

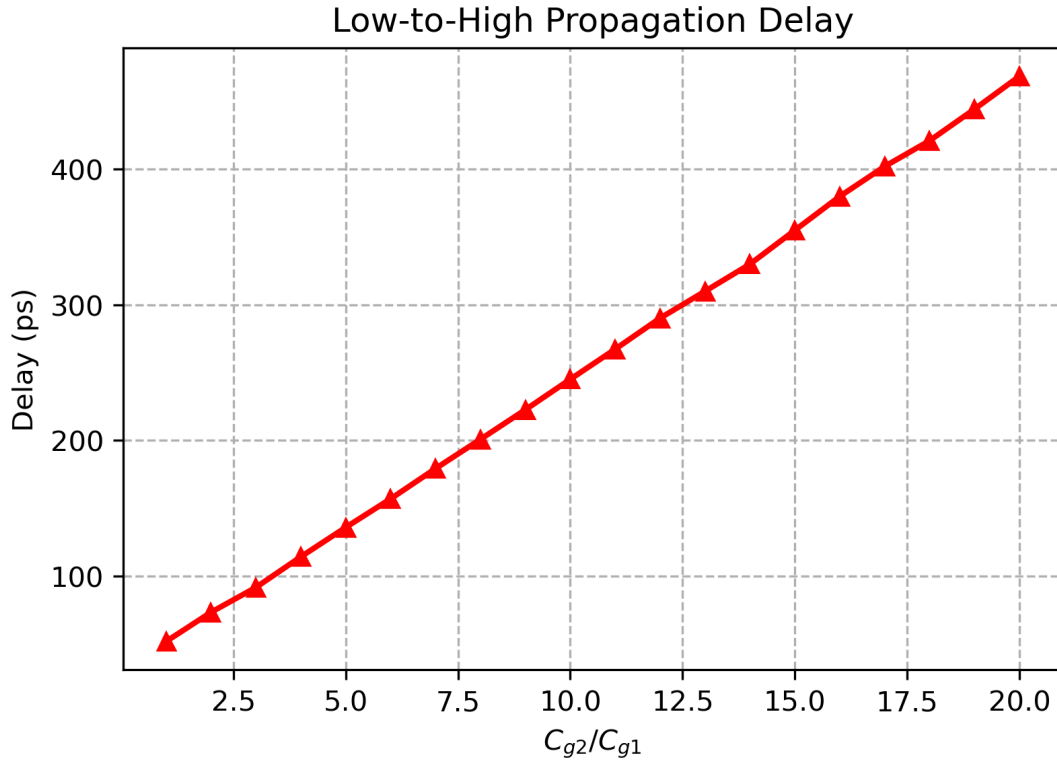


Figure 3: Low-to-High propagation delay (t_{pLH}) with respect to $\frac{C_{L(ext)}}{C_{in}}$

Exercise 4.1

The transient responses of the interconnect line modeled under varying segmentation degrees, $N = \{1, 2, 3, 4, 5\}$, are illustrated in Figure 4. A preliminary analysis of the global waveform plot reveals that, with the exception of the heavily under-sampled baseline configuration ($N = 1$ lumped RC network), the transient output curves of the remaining distributed ladder models appear nearly identical. However, by applying a localized zoom onto the propagation thresholds of the step response (as detailed in Figure 5), subtle propagation deviations among the higher-order networks can be systematically resolved.

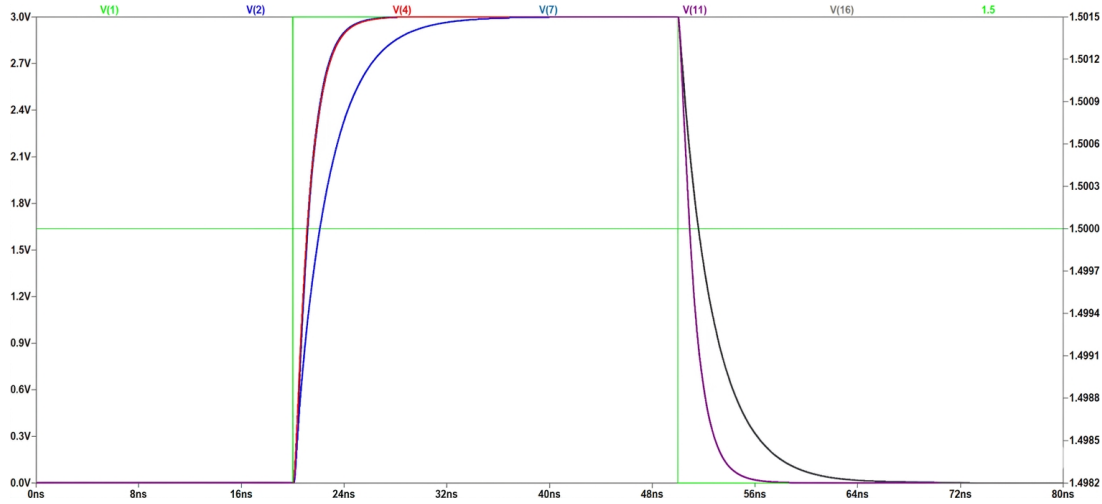


Figure 4: Transient responses of the interconnect line for $N=1,2,3,4,5$ segmentation degree

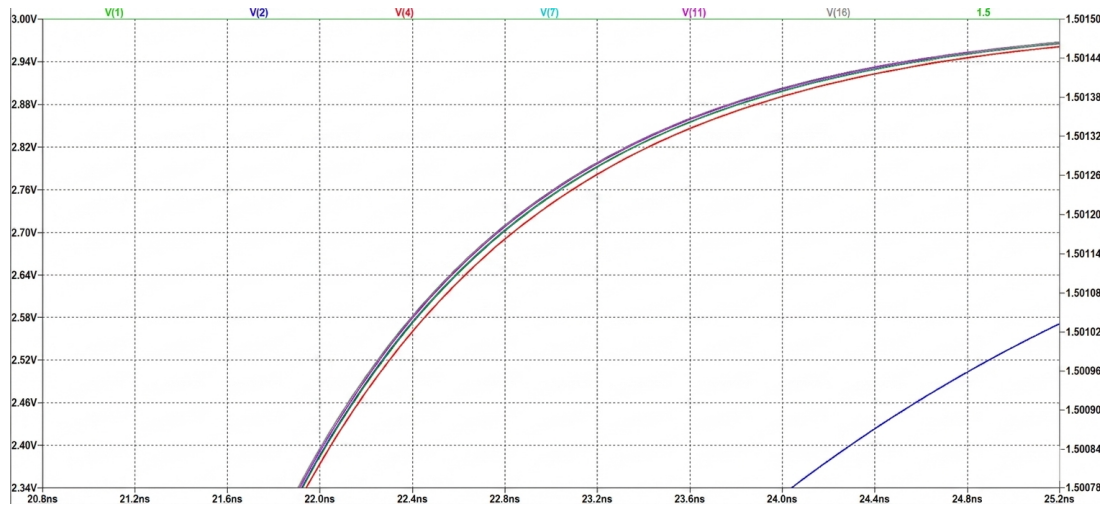


Figure 5: Localized zoom on figure 4

By closely analyzing the zoomed transient behavior, it is apparent that as the number of segments increases from $N = 2$ to $N = 5$, the output waveforms exhibit asymptotic convergence toward the true distributed physical behavior of the polysilicon interconnect line.

Quantitatively, the response deviation between the $N = 4$ and $N = 5$ distributed models is practically negligible, demonstrating that the higher-order network has reached a point of diminishing returns in terms of modeling accuracy. Dividing the interconnect wire into a fifth section introduces additional internal nodes and intermediate passive components (R and C elements) into the SPICE netlist. This escalates the circuit matrix size and increases the computational simulation time without yielding any meaningful refinement in the extracted delay parameters.

Consequently, from a qualitative design engineering perspective, the $N = 4$ configuration is explicitly identified as the optimal optimization boundary.

Exercise 5.1

Utilizing the LTspice simulator, the transient voltage waveforms are extracted and illustrated in Figure 6. Based on these dynamic characteristics, the propagation delays between the input

port of the third inverter and the output node of the distributed RC network (which serves as the driving input for the fourth inverter) are calculated. The numerical values for the high-to-low and low-to-high propagation delays are systematically estimated at $t_{pHL} = 746.75$ ps and $t_{pLH} = 660.17$ ps, respectively.

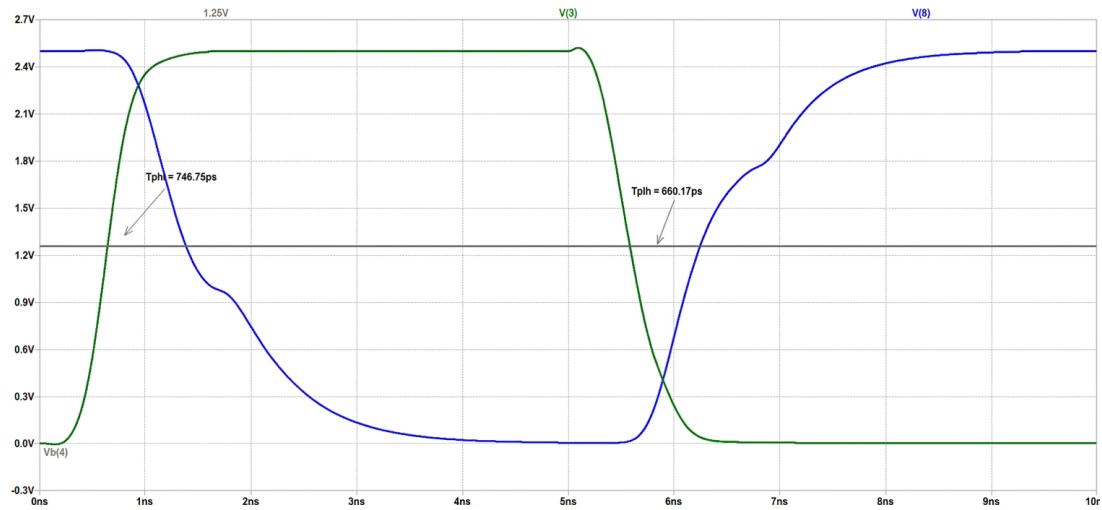


Figure 6: Transient voltage response comparing the input stream of the third inverter against the output node of the RC network (input port of the fourth inverter).